

The Two-Player $3n \pm 1$ Game: A Binary Normal Form and an Audit-Hardened Reduction

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<https://github.com/Grisha-Pochuev/3n-plus-minus-1-game>

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Abstract

We study the two-player $3n \pm 1$ game on positive odd integers. Earlier versions of this manuscript claimed a complete no-DRAW proof. External and adversarial audits found genuine defects in the global routing argument. The present version is deliberately claim-safe: it retains the binary normal form, constant-tail identities, factor-removal lemmas, source embedding, and the repaired factorful exponent-one entry, but it withdraws the unconditional global theorem until two explicit routing-provenance obligations are proved.

The first unresolved obligation is the seeded A/B_2 reset: after a bounded A -streak, a valuation-two B transfer may return to an obligation at a numerically unrelated source, and the proof must exhibit a certified strict proof-tree descendant before the control rank is reset. The second is the semantic completeness of the factor/high-return/marked-tail router: every actual DRAW continuation must be shown to fall into one of the declared constructors with its finite witness provenance intact. The formerly used anchor estimate is not assumed; the concrete stress test $20 \rightarrow 30 \rightarrow 45 \rightarrow 68$ with $B(68) = 25 > 20$ is included as a permanent negative regression. Thus this document is a rigorous reduction and repair dossier, not a claim that the prize problem has already been completely solved.

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1 The game and the theorem

The current state is a positive odd integer n . For $n > 1$ the player to move chooses one of

$$3n - 1, \quad 3n + 1,$$

and divides the chosen value by the largest possible power of 2, leaving an odd integer. If the resulting odd integer is 1, the player who made the move wins immediately. Players alternate with perfect information.

A position is called WIN for the player to move if at least one move leads to LOSS. It is called LOSS if every legal move leads to WIN. These two classes are the inductively generated finite-remoteness classes. Any position not in either class is called DRAW.

Target no-DRAW statement. Every positive odd starting position has finite remoteness; equivalently, the $3n \pm 1$ game has no DRAW positions and optimal play reaches 1 after finitely many moves. In this version this is the *target statement*, not an unconditional theorem. The exact remaining hypotheses needed for the conditional closure are stated in Section 10.

The theorem is not a statement about arbitrary play. There are legal cycles, such as

$$5 \longrightarrow 7 \longrightarrow 5,$$

under suitable choices by the players.

2 Binary conjugation

Write an odd state as $n = 2m + 1$ and define

$$F(m) = \left\lceil \frac{3m}{2} \right\rceil.$$

For a nonnegative integer x , let $R(x)$ be the integer obtained by deleting the maximal alternating suffix of the binary expansion of x . Thus $R(1001_2) = 10_2$ and $R(10101_2) = 0$.

Proposition 2.1 (First normal form). *For $m > 0$, the two children in m -coordinates are exactly*

$$F(m), \quad F(R(m)).$$

Proof. One has

$$3(2m + 1) - 1 = 2(3m + 1), \quad 3(2m + 1) + 1 = 2(3m + 2).$$

Exactly one of $3m + 1$ and $3m + 2$ is odd. In the other expression, each additional factor of two removed corresponds to crossing one additional alternating bit at the end of the binary expansion of m . The process stops at the first repeated adjacent bit, which is precisely the deletion defining $R(m)$. Returning to m -coordinates gives the displayed pair. $\square$

Since both children lie in the image of F , we represent the state $F(q)$ by q . The conjugated game has terminal state 0 and moves

$$A(q) = \left\lceil \frac{3q}{2} \right\rceil, \quad B(q) = R(A(q)).$$

Proposition 2.2 (Strict contraction). *For every $q > 0$,*

$$B(q) < q.$$

Proof. Deleting a nonempty binary suffix gives

$$B(q) \leq \left\lfloor \frac{A(q)}{2} \right\rfloor \leq \left\lfloor \frac{3q+1}{4} \right\rfloor < q$$

for $q \geq 2$, while $B(1) = 0$. □

Thus all unbounded difficulty comes from the expanding branch A and from the unbounded alternating suffix removed by B .

3 Constant-tail coordinates and coefficient sources

For a positive odd integer a , an exponent $r \geq 1$, and a phase $e \in \{0, 1\}$, put

$$Q_r^e(a) = a2^r - e.$$

This is a binary word whose final constant block has length r and consists of the bit e .

Lemma 3.1 (Long-tail recurrence). *For every positive odd a , phase e , and $r \geq 3$,*

$$A(Q_r^e(a)) = Q_{r-1}^e(3a), \quad B(Q_r^e(a)) = Q_{r-2}^e(3a).$$

Moreover

$$B(Q_1^e(a)) = B(Q_2^e(a)).$$

Proof. For $r \geq 3$, the expanding child still ends in at least two equal bits; hence its maximal alternating suffix consists only of the final bit. This gives both long-tail identities. At the boundary, the binary words of $3a - e$ and $6a - e$ differ by appending one bit that extends the same alternating suffix, so the same prefix remains after deletion. □

Define the embedded three-free coefficient

$$J(s) = 2A(s) + 1.$$

Every positive odd coefficient a has a unique factorization

$$a = 3^k J(s), \quad k \geq 0.$$

Indeed, remove all factors of 3 and call the remaining odd integer c . Then $c \not\equiv 0 \pmod{3}$ and $(c-1)/2$ is 0 or 2 modulo 3. The image of

$$A(2t) = 3t, \quad A(2t+1) = 3t+2$$

is exactly the set of nonnegative integers not congruent to 1 modulo 3, and A is injective. Hence there is a unique s with $A(s) = (c-1)/2$, equivalently $c = J(s)$. Unique factorization supplies the unique exponent k . The integer s is called the coefficient source of a and is denoted $\text{src}(a)$.

Lemma 3.2 (Source boundary). *For every $s > 0$ there is a phase*

$$\alpha(s) = 1 - \left(\left\lfloor \frac{s}{2} \right\rfloor \bmod 2 \right)$$

such that

$$\text{odd}(3J(s) + 1 - 2\alpha(s)) = J(A(s))$$

with 2-adic valuation one, while the opposite phase selects $J(B(s))$ with valuation at least two.

Proof. Put $u = A(s)$. Then $J(s) = 2u + 1$ and

$$3J(s) + 1 - 2e = 2(3u + 2 - e).$$

The bracket is odd exactly when $e = 1 - (u \bmod 2)$. For $s = 2t$ or $2t + 1$ one has $u \bmod 2 = t \bmod 2$, so this phase is exactly $e = \alpha(s)$. If u is even, then $e = 1$ and $3u + 1 = J(u)$; if u is odd, then $e = 0$ and $3u + 2 = J(u)$. Thus the valuation-one child is $J(A(s))$. The opposite sign is the other original $3n \pm 1$ child of the odd state $J(s)$; under the conjugacy of Proposition 2.1 that child is $J(B(s))$. Since it is not the valuation-one branch, its valuation is at least two. $\square$

Multiplying a constant-tail coefficient by 3 preserves its coefficient source. Passing through a factor-free signed boundary with coefficient $J(s)$ therefore exposes exactly one ordinary A/B move on the source.

For a positive state q , write $\kappa(q)$ for the odd coefficient in its canonical constant-tail coordinates.

Lemma 3.3 (Coefficient pullback). *For every positive odd w ,*

$$\kappa(3w) = J(R(w)), \quad \kappa(3w - 1) = J(R(2w - 1)).$$

Proof. Let $m \geq 1$ be the length of the maximal alternating suffix of the odd word w , and put $z = R(w)$. If $z > 0$, maximality says that the leading bit of the suffix equals the last bit of z . Because the suffix ends in 1, its value is

$$t_m = \begin{cases} (2^m - 1)/3, & m \text{ even}, \\ (2^{m+1} - 1)/3, & m \text{ odd}. \end{cases}$$

Thus $w = 2^m z + t_m$. If m is even, z is even and

$$3w + 1 = 2^m(3z + 1) = 2^m J(z).$$

If m is odd, z is odd and

$$3w + 1 = 2^m(3z + 2) = 2^m J(z).$$

If $z = 0$, the whole word is alternating; then m is odd and $3w + 1 = 2^{m+1}$, whose odd part is $J(0) = 1$. Hence in all cases $\kappa(3w) = \text{odd}(3w + 1) = J(R(w))$.

For the second identity set $v = 2w - 1$, which is odd. Since $3v + 1 = 2(3w - 1)$, the first identity applied to v gives

$$\kappa(3w - 1) = \text{odd}(3w - 1) = \text{odd}(3v + 1) = J(R(v)) = J(R(2w - 1)).$$

$\square$

4 Finite outcomes, witnesses, and ranks

The terminal state 0 is LOSS with height $h(0) = 0$. A finite WIN state has height

$$h(x) = 1 + \min\{h(y) : y \text{ is a LOSS child of } x\},$$

and a finite LOSS state has height

$$h(x) = 1 + \max\{h(y) : y \text{ is a child of } x\}.$$

A state whose outcome has been proved but which is being carried only to justify a later incidence is called a *routing witness*. A *retained proof token* is a finite state for which the proof has also established the comparison needed by the rank: it either seeds an empty inner token set at a one-shot seed transition, or replaces a previously retained token by certified descendants of strictly smaller height. This distinction is important: a newly discovered finite state is never silently appended to a ranked multiset.

Lemma 4.1 (Multiset token rank). *For a finite multiset M of proof heights define*

$$\Phi(M) = \#_{h \in M} \omega^h,$$

the natural ordinal sum of the monomials ω^h . Replacing one token of height H by any finite multiset of certified descendants of heights strictly below H strictly decreases Φ .

Proof. In Cantor normal form the coefficient of ω^H decreases by one, while all inserted terms have lower exponent. No finite collection of lower terms can compensate for the loss at exponent H . $\square$

Every positive state q has unique canonical constant-tail coordinates $q = Q_r^e(a)$ with a odd and $r \geq 1$. If $a = 3^k J(s)$, define the *state source* $\rho(q) = s$.

Lemma 4.2 (Universal state-source bound). *For every $q > 0$,*

$$\rho(q) \leq \frac{q-1}{6}.$$

Proof. Write $q = Q_r^e(3^k J(s))$. Since $r \geq 1$ and $e \leq 1$, $q \geq 2J(s) - 1$. Moreover $J(s) = 2\lceil 3s/2 \rceil + 1 \geq 3s + 1$. Hence $q \geq 6s + 1$. $\square$

Lemma 4.3 (Boundary coefficient descent). *Let a be positive and odd and*

$$p = B(Q_1^e(a)) = B(Q_2^e(a)).$$

If $p > 0$, then its canonical coefficient satisfies

$$\kappa(p) \leq \frac{3a+1}{4}.$$

In particular $\kappa(p) < a$ for $a \geq 3$.

Proof. The common child is $p = R(3a - e)$. Deleting a nonempty suffix gives $p \leq \lfloor (3a - e)/2 \rfloor$. For every positive integer y , the odd coefficient in its canonical constant-tail coordinates is at most $(y+1)/2$. Therefore

$$\kappa(p) \leq \frac{p+1}{2} \leq \frac{3a+1}{4}.$$

The final inequality is strict relative to a for $a \geq 3$. $\square$

5 Side identities and typed boundary objects

The finite routing system uses two elementary binary identities. We record them here because they are the only places where an exceptional residue class enters.

Lemma 5.1 (Ordinary side relation). *Put $S(q) = B(q)$ and $T(q) = B(A(q))$. If*

$$q \not\equiv 1, 3, 12, 14 \pmod{16},$$

then

$$T(q) \in \{A(S(q)), B(S(q))\}.$$

If q is in one of the four exceptional classes, then $A(q)$ is not exceptional.

Proof. Let $x = A(q)$ and let k be the length of the maximal alternating suffix of x . Direct reduction of $A(q)$ modulo 8 shows that $k \geq 3$ is possible exactly for $q \equiv 1, 3, 12, 14 \pmod{16}$. Outside these classes $k = 1$ or 2 . Writing x as $S(q)$ followed by that one- or two-bit alternating word, the formulas

$$A(2u) = 3u, \quad A(2u + 1) = 3u + 2$$

show that $A(x)$ is obtained from $A(S(q))$ by appending the corresponding one- or two-bit alternating word. Deleting its maximal alternating suffix therefore leaves either $A(S(q))$ or $B(S(q))$. The four exceptional residue classes map under A to $2, 5, 10, 13 \pmod{16}$, none of which is exceptional. $\square$

Corollary 5.2 (No four consecutive wins). *For no q can all four states*

$$q, A(q), A^2(q), A^3(q)$$

be WIN.

Proof. If q and $A(q)$ are both WIN, then $B(q)$ must be LOSS; the same holds at the next two positions. If q is ordinary, Lemma 5.1 makes $B(A(q))$ a child of the LOSS state $B(q)$, impossible. Hence q must be exceptional. Applying the same argument to $A(q)$ forces $A(q)$ exceptional, contrary to Lemma 5.1. $\square$

Lemma 5.3 (Alternating lift identity). *For every positive integer z there are $m \geq 1$ and $\delta = 1 - (z \bmod 2)$ such that*

$$3z + 1 = Q_m^\delta(J(R(z))).$$

If $R(z) > 0$, one may take m equal to the length of the maximal alternating suffix of z ; if $R(z) = 0$, one takes one more than the binary length of z .

Proof. First suppose $r = R(z) > 0$, let k be the alternating-suffix length and put $\epsilon = r \bmod 2$. Then

$$z = 2^k r + t_{k,\epsilon}, \quad t_{k,0} = \left\lfloor \frac{2^k}{3} \right\rfloor, \quad t_{k,1} = \left\lfloor \frac{2^{k+1}}{3} \right\rfloor.$$

Since $J(r) = 3r + 1 + \epsilon$, direct substitution gives

$$3t_{k,\epsilon} + 1 + \delta = 2^k(1 + \epsilon),$$

the two parity cases being exactly the two remainders of a power of 2 modulo 3. Hence

$$3z + 1 + \delta = 2^k J(r),$$

which is the required identity with $m = k$. If $R(z) = 0$, the whole binary word of z is alternating and starts with 1, so $z = t_{k,1}$. The same calculation gives $3z + 1 + \delta = 2^{k+1} = 2^{k+1} J(0)$. $\square$

Lemma 5.4 (Exceptional side attachment). *Let q be exceptional and suppose*

$$q \text{ is WIN, } \quad A(q) \text{ is DRAW, } \quad \ell = B(q) \text{ is LOSS.}$$

Then for some $m \geq 1$ and $\delta \in \{0, 1\}$ the two children of $A(q)$ are exactly

$$Q_m^\delta(J(\ell)), \quad Q_{m+1}^\delta(J(\ell)),$$

and this adjacent pair contains a continuing DRAW.

Proof. Write $q = 16t + c$ with $c \in \{1, 3, 12, 14\}$. The four direct exceptional formulas are

c	ℓ	$B(A(q))$	$A^2(q)$
1	$R(6t)$	$18t + 1$	$36t + 3$
3	$R(6t + 1)$	$18t + 4$	$36t + 8$
12	$R(6t + 4)$	$18t + 13$	$36t + 27$
14	$R(6t + 5)$	$18t + 16$	$36t + 32$.

In every row put respectively $z = 6t, 6t + 1, 6t + 4, 6t + 5$. Then $\ell = R(z)$ and $B(A(q)) = 3z + 1$. By Lemma 5.3,

$$B(A(q)) = Q_m^\delta(J(\ell))$$

for some $m \geq 1$. The last column is $2B(A(q)) + \delta$, so it equals $Q_{m+1}^\delta(J(\ell))$. These are precisely the two children of the DRAW state $A(q)$; therefore neither is LOSS and at least one is DRAW. $\square$

For a source x and phase e , let $O(x, e)$ denote the following exact *DRAW obligation*:

- (a) $Q_1^e(J(x))$ is DRAW; or
- (b) there is a DRAW state whose two children are exactly $Q_2^e(J(x))$ and $Q_1^e(J(x))$.

This definition introduces no outcome assumption beyond the displayed DRAW state.

Lemma 5.5 (Hidden parent for a typed three/two frame). *Let a be positive, odd and not divisible by 3, and suppose*

$$a \equiv 1 + e \pmod{3}.$$

Put $H = Q_3^e(a)$, $G = Q_2^e(a)$ and

$$K = \frac{2H + e - 1}{3}.$$

Then K is an integer and

$$\text{moves}(K) = \{H, G\}.$$

Proof. For $e = 0$ the congruence gives $H \equiv 2 \pmod{3}$ and $K = (2H - 1)/3$; for $e = 1$ it gives $H \equiv 0 \pmod{3}$ and $K = 2H/3$. These are exactly the two inverse branches of the injective map $A(q) = \lceil 3q/2 \rceil$. The last three bits of H are equal, so the contracting child removes only the last bit and equals G . $\square$

6 Fixed-tail factor removal and entry typing

The next two lemmas isolate the only raw entry phenomenon that is not already an obligation or a factor frame.

Lemma 6.1 (Fixed-tail factor-removal diamond). *Let c be positive and odd, $e \in \{0, 1\}$, and $r \geq 2$. Put*

$$X = Q_r^e(3c), \quad G = Q_r^e(c), \quad H = Q_{r+1}^e(c), \quad Y = A(G).$$

Then

$$\text{moves}(H) = \{X, Y\}.$$

If X is DRAW, either G is DRAW, or the diamond determines a finite routing witness together with its exact incidence to G, H, X, Y and the other child of G .

Proof. Since $r + 1 \geq 3$, Lemma 3.1 gives $A(H) = X$. For $r \geq 3$ it also gives $B(H) = Q_{r-1}^e(3c) = A(G)$; for $r = 2$ the boundary identity gives the same equality. Thus $B(H) = Y$.

Because X is DRAW, H is not LOSS. If H is WIN, then Y is a LOSS child of H . Assume H is DRAW. Then Y is not LOSS. If G is LOSS, then Y is WIN. If G is WIN, either Y itself is a LOSS witness, or the other child of G is a LOSS witness. These are all outcome orientations. In every finite branch the witness and the parent-child incidence just displayed are known exactly; no token-rank comparison is asserted at this stage. $\square$

Definition 6.2 (Entry controls). *A boundary entry is an obligation $O(x, e)$ or, more generally, a factor-free adjacent frame*

$$\{Q_{m+1}^e(J(x)), Q_m^e(J(x))\}, \quad m \geq 1,$$

known to contain a continuing DRAW. Its remaining tail length is recorded as τ . The typed three/two frame satisfying Lemma 5.5 is a distinguished $m = 2$ boundary constructor. A loss-sibling entry is one of the finitely many local constructors in which a continuing DRAW is accompanied by a finite WIN/LOSS routing witness with a proved parent/child incidence: the fixed-tail diamond of Lemma 6.1, the exponent-one predecessor diamond of Lemma 6.4, an ordinary side diamond of Lemma 5.1, the exceptional side attachment of Lemma 5.4, or the hidden-parent frame of Lemma 5.5. The constructor records its remaining constant-tail/factor exponent as the local counter τ .

Lemma 6.3 (Entry typing). *Every finite branch produced by Lemma 6.1 is a loss-sibling entry in the sense of Definition 6.2; no ranked multiset is changed in order to make this classification.*

Proof. The proof of Lemma 6.1 gives the finite witness and its exact incidence to G, H, X, Y and the other child of G . These are precisely the data of the first loss-sibling constructor. Classification is therefore immediate; subsequent tail/factor normalization belongs to the typed routing relation and is ranked there by τ and, when available, by descendant-token comparisons. $\square$

Lemma 6.4 (Exponent-one lift). *Let a be positive and odd and $k \geq 1$. If $R_k = Q_1^e(3^k a)$ is DRAW, then*

$$P_{k-1} = Q_2^e(3^{k-1} a)$$

is either DRAW or WIN; in the latter case its other child is a finite LOSS witness.

Proof. The boundary recurrence gives $A(P_{k-1}) = R_k$. A state with a DRAW child cannot be LOSS. If P_{k-1} is WIN, the DRAW child cannot witness the win, so its other child is LOSS. $\square$

Proposition 6.5 (Arbitrary factorful exponent-one entry). *Let $a = J(s)$ and $k > 0$. If $Q_1^e(3^k a)$ is DRAW, then after finitely many outcome-compatible local steps one obtains either a typed entry of Lemma 6.3, or the factor-free exponent-two state*

$$Q_2^e(a) \text{ DRAW.}$$

Proof. Apply Lemma 6.4. Its finite branch is a loss-sibling entry by Definition 6.2. Otherwise $Q_2^e(3^{k-1}a)$ is DRAW. Apply Lemma 6.1 with $r = 2$ repeatedly. Whenever its finite branch occurs, Lemma 6.3 supplies a typed entry. If it never occurs, one factor of 3 is removed at the same tail exponent at each step, so after $k - 1$ steps the displayed factor-free exponent-two state is reached. $\square$

Proposition 6.6 (Complete exponent-one/two boundary reduction). *Let $s \geq 0$, $k \geq 0$, $e \in \{0, 1\}$, and let*

$$Q_r^e(3^k J(s)), \quad r \in \{1, 2\},$$

be DRAW. For $s > 0$, after finitely many local steps at least one of the following occurs: a typed entry is obtained, the retained source strictly decreases, or the factor-free exponent-two state $Q_2^e(J(s))$ is DRAW. For $s = 0$ the corresponding statement is Lemma 8.1.

Proof. Assume $s > 0$. There are four cases. If $r = 1, k = 0$, the state is the first alternative of the obligation $O(s, e)$ and is already a boundary entry. If $r = 1, k > 0$, use Proposition 6.5. If $r = 2, k = 0$, we are at the displayed factor-free exponent-two state. Finally, if $r = 2, k > 0$, apply Lemma 6.1 repeatedly with $r = 2$: a finite branch is a loss-sibling entry by Definition 6.2, while otherwise one factor of 3 is removed at the same exponent at each step and after k steps the factor-free exponent-two state is reached. These cases are disjoint and exhaustive. $\square$

7 Factor-free exponent-two base entry

Let $x > 0$, $a = J(x)$ and

$$P = Q_1^e(a), \quad U = Q_2^e(a), \quad b = B(P) = B(U), \quad F = A(U) = Q_1^e(3a), \quad g = 1 - e.$$

Assume U is DRAW and x is the least coefficient source of any DRAW. This is a *factor-free* coefficient $J(x)$. The common child b cannot be 0, because 0 is LOSS and a DRAW has no LOSS child. By Lemma 4.3, the canonical coefficient of b is strictly below $J(x)$. After deleting any factor of 3 its three-free coefficient is smaller still, hence it equals $J(t)$ with $t < x$ because J is strictly increasing. Therefore the state source of b is below x , and b is not DRAW. No such inference is made for a coefficient $3^k J(x)$ with $k > 0$. As a child of a DRAW it is not LOSS, so

$$b \text{ is WIN,} \quad F \text{ is DRAW.}$$

Lemma 7.1 (First-factor anchor identity). *For the states above,*

$$9J(x) + 1 - 2e = 2^v J(b), \quad v \geq 1. \tag{1}$$

Proof. Put $a = J(x)$. If $e = 0$, then $P = 2a$, $A(P) = 3a$ and $b = R(3a)$. Apply the first identity of Lemma 3.3 to the odd integer $w = 3a$:

$$\text{odd}(9a + 1) = \kappa(9a) = J(R(3a)) = J(b).$$

If $e = 1$, then $P = 2a - 1$, $A(P) = 3a - 1$ and $b = R(3a - 1)$. The second identity of Lemma 3.3, again with $w = 3a$, gives

$$\text{odd}(9a - 1) = J(R(6a - 1)).$$

By the boundary identity of Lemma 3.1, $R(6a - 1) = R(3a - 1) = b$. Hence in both phases the odd part of $9a + 1 - 2e$ is exactly $J(b)$, which proves (1). $\square$

Proposition 7.2 (Base-entry attachment). *Every factor-free exponent-two DRAW at a globally minimal source has a finite continuation which either strictly lowers the retained source or produces a boundary/loss-sibling typed entry.*

Proof. Suppose first that $e = 1 - \alpha(x)$. The first source-boundary numerator is divisible by 4, while

$$9J(x) + 1 - 2e = 3(3J(x) + 1 - 2e) - 2(1 - 2e) \equiv 2 \pmod{4}.$$

Thus $v = 1$. Writing $u = A(x)$ gives $b = 3u + 1$. The two children of the DRAW state F are

$$T = Q_1^g(J(b)), \quad C = R(T),$$

and direct appended-bit deletion gives $C \in \{A(b), B(b)\}$. If T is DRAW, this is the boundary entry $O(b, g)$. If C is DRAW, the other child ℓ of the WIN state b is a finite LOSS witness. If b is ordinary, this is the ordinary-side loss-sibling constructor. If b is exceptional and $C = A(b)$, Lemma 5.4 gives an adjacent DRAW frame over ℓ , hence a boundary/loss-sibling entry. It remains only the exceptional orientation $C = B(b)$. In that orientation $b \leq (9x + 5)/2$ and the alternating suffix of $A(b)$ has length at least three, so

$$C = B(b) \leq \frac{A(b)}{8} \leq \frac{3b + 1}{16}.$$

Lemma 4.2 then yields

$$\rho(C) \leq \frac{C - 1}{6} < \frac{3b + 1}{96} \leq \frac{27x + 17}{192} < x,$$

a strict retained-source exit.

Now suppose $e = \alpha(x)$. Lemma 3.2 gives $3J(x) + 1 - 2e = 2J(A(x))$, hence $v \geq 2$. If $v \geq 4$, then

$$16J(b) \leq 9J(x) + 1 \leq 27x + 19,$$

and $J(b) \geq 3b + 1$ implies

$$b \leq \frac{9x + 1}{16} < x.$$

Thus the continuing DRAW has smaller retained source. If $v = 2$, the two children of F are exactly $Q_2^g(J(b))$ and $Q_1^g(J(b))$, which is the second form of $O(b, g)$. If $v = 3$, they are the typed three/two frame $Q_3^g(J(b)), Q_2^g(J(b))$. Reducing (1) modulo 3 gives

$$J(b) \equiv 1 + g \pmod{3},$$

so Lemma 5.5 applies. The hidden parent is a boundary entry; its finite side, if one is exposed, is a loss-sibling entry. Hence no untyped base row remains. $\square$

8 The source-zero branch

For $k \geq 0$, $r \geq 1$ and $e \in \{0, 1\}$ put

$$S_{k,r}^e = Q_r^e(3^k).$$

These are precisely the constant-tail states with coefficient source zero.

Lemma 8.1 (Zero-source boundary normalization). *If a source-zero DRAW exists, then after finitely many outcome-compatible moves one reaches a typed boundary or loss-sibling entry. No ranked token is installed and no control reset is used before this finite normalization is complete.*

Proof. For $r \geq 3$, Lemma 3.1 gives

$$\text{moves}(S_{k,r}^e) = \{S_{k+1,r-1}^e, S_{k+1,r-2}^e\}.$$

Following a DRAW child strictly lowers r , so a DRAW with $r \in \{1, 2\}$ is reached after finitely many moves. If $k = 0$ already at that boundary, the four possibilities are explicit:

$$Q_1^1(1) = 1 \text{ WIN}, \quad Q_2^1(1) = 3 \text{ WIN}, \quad Q_1^0(1) = 2 \text{ LOSS}, \quad Q_2^0(1) = 4 \text{ LOSS}.$$

Hence a boundary DRAW must have accumulated at least one factor of 3.

If the reached boundary has tail exponent one, put $n = k + 1$. If it has tail exponent two, its children are the common raw child and $S_{k+1,1}^e$. If the raw child is DRAW, again put $n = k + 1$; otherwise the raw child is WIN and forces $S_{k+1,1}^e$ to be DRAW, in which case put $n = k + 2$. Thus in all cases one obtains a boundary DRAW configuration whose raw/signed exits are governed, for some $n \geq 1$, by

$$3^n + 1 - 2e = 2^j J(y).$$

The exact 2-adic valuation is

	n odd	n even
$e = 0$	$j = 2$	$j = 1$
$e = 1$	$j = 1$	$j = 2 + v_2(n)$.

For $j \geq 2$ the raw and signed exits form the adjacent factor-free frame

$$Q_{j-1}^{1-e}(J(y)), \quad Q_j^{1-e}(J(y)),$$

containing a continuing DRAW; by Definition 6.2 this is a boundary entry, with its finite tail length recorded in τ . For $j = 1$ one has

$$y = \frac{3^{n-1} - 1}{2}.$$

When $e = 0$ the raw exit is $A(y)$; when $e = 1$ it is $B(y)$. The raw and signed states are the two exits of the same boundary DRAW configuration. If the signed exit is DRAW, it is already the exponent-one form of an obligation. If the raw exit is the continuing DRAW and the signed exit is finite, that finite state is necessarily WIN; choosing a canonical LOSS child of it supplies an exact finite routing witness, hence a loss-sibling entry. If both exits are DRAW, use the signed obligation. Thus every row reaches a typed entry after a finite pre-entry computation; no impossible “source below zero” comparison is used. $\square$

Lemma 8.2 (*A-selecting side diamond*). *Let $x > 0$, $e = \alpha(x)$, $w = Q_1^e(J(x))$, and $y = A(x)$. Then*

$$B(w) \in \{A(y), B(y)\}.$$

Proof. The source-boundary identity gives $A(w) = Q_1^{1-e}(J(y)) = 4A(y) + 1 + e$. For $e = 0$ the appended bits are 01; for $e = 1$ they are 10. If the first appended bit repeats the last bit of $A(y)$, deleting the alternating suffix leaves $A(y)$; otherwise the deletion continues through the maximal alternating suffix of $A(y)$ and leaves $B(y)$. $\square$

Lemma 8.3 (*Universal B-transfer diamond*). *Let $x > 0$, let $e = 1 - \alpha(x)$, put $a = J(x)$ and $g = 1 - e$, and factor*

$$3a + 1 - 2e = 2^j J(y), \quad j \geq 2,$$

so $y = B(x)$. Define

$$P = Q_1^e(a), \quad U = Q_2^e(a), \quad D = Q_j^g(J(y)), \quad C = Q_{j-1}^g(J(y)), \quad F = A(U).$$

Then $D = A(P)$ and $C = B(P) = B(U)$. If X is the common child of D, C and Y is the other child of C , then

$$B(F) = Y.$$

Consequently an obligation in the B-selecting phase transfers a continuing DRAW to the adjacent frame $\{D, C\}$; when $j = 2$ the transferred frame is again an obligation in the opposite phase.

Proof. The formulas for D and C are the signed boundary formula and the shared-child identity. The remaining equality is a direct substitution in the three cases $j = 2$, $j = 3$, and $j \geq 4$: the constant suffix of D determines its common child X , while the one-step longer factor state F has the same raw side Y . For the outcome assertion, a common child of a DRAW cannot be LOSS. If the exponent-one member is DRAW, the frame $\{D, C\}$ already contains a DRAW. In the parent form of an obligation, the only remaining orientation has the exponent-two member DRAW; if C were WIN, outcome recursion through the displayed diamond would force Y both LOSS and a child of the DRAW state F , impossible. Hence C is DRAW. For $j = 2$, D, C are exactly the exponent-two/one pair over $J(y)$. $\square$

9 Audit-hardened routing reduction

The preceding sections are local arithmetic and outcome lemmas. The remaining issue is not another binary identity; it is the global bookkeeping needed to compose the local diamonds without silently installing an unrelated proof height or silently changing a ranked source anchor.

9.1 The old anchor shortcut is permanently withdrawn

The earlier routing proof attached a retained integer p to an A -streak and, at a later B -selecting cursor y , used an estimate of the form

$$y \leq \frac{3p + 1}{2}$$

to conclude that a high-valuation B step lowered the retained source. That provenance is false after more than one canonical A continuation.

Lemma 9.1 (Anchor stress test). *Let $x_0 = 20$ and $x_{i+1} = A(x_i)$. Then*

$$x_0, x_1, x_2, x_3 = 20, 30, 45, 68.$$

The canonical phases remain A -selecting through the first three sources and switch at 68. The selected B transition at 68 has valuation 3 and

$$B(68) = 25 > 20.$$

Consequently no proof may infer a decrease relative to the original anchor 20 merely from the later high-valuation B transfer.

Proof. Directly,

$$A(20) = 30, \quad A(30) = 45, \quad A(45) = 68,$$

and

$$\alpha(20), \alpha(30), \alpha(45), \alpha(68) = 1, 0, 1, 1.$$

The lifted phase flips at every canonical A continuation, so the required phase at 68 is 0, opposite to $\alpha(68) = 1$. Moreover

$$3J(68) + 1 = 616 = 2^3 \cdot 77, \quad 77 = J(25),$$

which gives the asserted valuation and $B(68) = 25$. □

The revised routing architecture therefore treats the displayed integers in an A -streak as *temporary cursors*. A cursor is never promoted to a ranked source merely because it is smaller than a later cursor. A control reset may be justified only by one of the two certificates below:

- (R1) an actual DRAW state whose retained coefficient source is proved smaller than the currently ranked source by an exact factor-free/source lemma;
- (R2) a finite WIN/LOSS proof token that is proved to be a proper proof-tree descendant of a token already present in the ranked multiset (or the unique initial token installation).

9.2 Finite control is not semantic completeness

The finite control vocabulary consists of the following names:

boundary_entry	loss_sibling_entry	a_obligation
a_test_4	a_test_3	a_test_2
a_test_1	b_select	b2_first
b2_ready	factor_fork	high_return
marked_tail	short_lift	terminal_macro.

The finite graph of these names is only a bookkeeping device. A graph edge is admissible in the mathematical proof only when its guard is derived from the actual game states and its finite witnesses have the exact parent/child incidence claimed for them. The accompanying JSON file therefore calls its edges *declared routing edges*, not proved game transitions.

The arithmetic part of a B -selecting transfer is safe without any retained anchor comparison. Lemma 8.3 gives an adjacent frame over the actual selected source $z = B(y)$. If the selected valuation is 2, this frame is again an exact obligation. If the valuation is at least 3, it is routed to the factor-frame normalizer. In neither row do we assert that z is below an older pre-streak source.

10 The two remaining proof obligations

The following statements are deliberately separated from the proved lemmas. They are the only assertions needed to turn the present reduction back into an unconditional no-DRAW proof, and they are *not* claimed proved in this version.

Obligation P1: seeded A/B_2 reset provenance. Suppose the inner routing layer already contains a ranked finite proof token. Follow an actual DRAW through any finite canonical A -streak and then through a valuation-two B transfer that returns to an obligation. Before the control is reset to a new `a_obligation`, one must exhibit either (i) a genuine ranked source decrease, or (ii) a finite witness which is a strict proof-tree descendant of an already ranked token. The statement must include multi- A streaks and all four exceptional residue classes. It is not enough that the new source is WIN, and it is not enough that it is smaller than the immediate cursor.

Obligation P2: semantic exhaustiveness of the factor router. For every actual DRAW entering any factor-router control, the complete outcome split must be derived from the game relation. The relevant controls are `factor_fork`, `high_return`, `marked_tail`, `short_lift`, and `terminal_macro`. Every case must land either in a declared non-reset routing edge or in a strict source/token edge. In particular, every reset to a higher control must carry the explicit descendant certificate for the token it replaces. No finite endpoint of unrelated height may be inserted merely because it is WIN or LOSS.

The long proof ledger in the repository contains many of the local identities needed for P1 and P2, including the factor-fork and high-return descendants. They are useful evidence, but the present manuscript does not promote the two global composition statements to theorems until their guards and provenance are checked as one closed transition system.

Proposition 10.1 (Conditional routing closure). *Assume P1 and P2. Then the marked routing relation admits a well-founded lexicographic/multiset ranking in which every control reset is preceded by a strict retained source or certified proof-token decrease, while all remaining routing is finite by the bounded control graph and its explicit tail/factor counters.*

Proof. The proof is the standard well-founded-fibre assembly. Rank finite proof witnesses by the natural multiset sum

$$\Phi(M) = \#_{h \in M} \omega^h$$

from Lemma 4.1. A certified descendant replacement strictly lowers Φ . A genuine retained-source edge lowers the numerical source and may reset later data. At fixed retained source and token multiset, P1 and P2 say exactly that no edge which resets the finite control is unranked. Delete the strict source/token edges. What remains is the declared finite control graph together with same-control recurrences whose explicit natural tail or factor counter decreases. The residual relation is well-founded. The well-founded-fibre lemma then gives the full marked relation. $\square$

11 Conditional no-DRAW consequence and present status

Theorem 11.1 (Conditional no-DRAW theorem). *If Obligations P1 and P2 hold, then the target no-DRAW statement of Section 1 follows.*

Proof. Assume a DRAW exists and choose a least coefficient source among DRAW positions. The long-tail reduction, the four explicit exponent-one/two boundary cases of Proposition 6.6,

and the factor-free base entry of Proposition 7.2 yield either a strict smaller-source DRAW (contradicting minimality) or a typed entry. The factorful exponent-one row is handled by Proposition 6.5; no reverse-factor lemma is used at exponent one and no coefficient/source shortcut is used with a factor 3^k .

Under P1 and P2, Proposition 10.1 makes every outcome-compatible marked continuation well-founded. But every DRAW has no LOSS child and at least one DRAW child, so the local outcome diamonds could be iterated forever while retaining an actual DRAW member. This contradicts well-foundedness. Hence no DRAW exists in the conjugated game, and Proposition 2.1 transfers finite remoteness to every original positive odd start. $\square$

Status. The implication above is proved; P1 and P2 are not. Accordingly version 5.0 **does not claim that the prize problem is solved**. This wording is intentional. It prevents a finite control certificate, a regression test, or a moving repository supplement from being mistaken for the missing semantic proof.

12 Verification boundary and reproducibility

The package contains three kinds of checks.

- `verify_v5_0.py` checks the exact arithmetic identities used in the proved part, the old source/coefficient counterexample, the permanent $20 \rightarrow 30 \rightarrow 45 \rightarrow 68$ anchor counterexample, and the declared finite control graph.
- `routing_certificate_v5_0.json` records controls and guards. Every edge is labelled either `pure-routing`, `strict-source`, `strict-token`, or `OPEN-P1/P2`. The verifier rejects an unlabelled reset edge.
- `OPEN_PROOF_OBLIGATIONS_v5_0.md` gives the exact mathematical targets still required for an unconditional theorem.

These checks are deliberately narrower than a formal proof. No end-to-end Lean formalization is claimed.

A Permanent adversarial regression cases

The following two examples are normative negative tests for every future revision.

Coefficient/source separation. For $s = 1$, $J(1) = 5$, $a = 3J(1) = 15$, and $e = 1$, the common boundary child is $R(44) = 22$. Its canonical coefficient is $11 < 15$, but $11 = J(3)$, so the coefficient source grows from 1 to 3. Therefore a smaller canonical coefficient does not imply a smaller source when powers of three are present.

Pre-streak anchor separation. The calculation of Lemma 9.1 gives $20 \rightarrow 30 \rightarrow 45 \rightarrow 68$ and $B(68) = 25 > 20$. Therefore a high-valuation B transfer after several A steps does not by itself lower the source retained at the beginning of the streak.

B Declared routing inventory

The table is a checklist, not a proof of P1/P2.

Control	Allowed non-strict successor classes
boundary_entry	finite tail countdown; then a_obligation or factor_fork
loss_sibling_entry	finite tail countdown; then factor_fork
a_obligation	bounded A test or factor fork; a smaller-source DRAW may re-enter only through a proved source comparison
a_test_4,3,2,1	next lower test or b_select
b_select	valuation 2 to b2_first ; valuation ≥ 3 to factor_fork ; no retained-anchor descent is asserted
b2_first	b2_ready
b2_ready	reset requires P1
factor_fork	high_return or strict certified exit
high_return	reset requires the P2 descendant certificate
marked_tail	shorter tail/short lift, or strict P2 token exit
short_lift	a_obligation
terminal_macro	every reset requires a P2 token certificate

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